Differential Geometry

Exercise 5

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5.1 Problem

Solve problem 6, page 78, of "Foundations of Lie Groups and Differentiable Manifolds", Warner. Let X, Y be vector fields on M, with $p \in M$. Let θ, ψ be the flows of X, Y, and define

$$\beta(t) = \psi_{-\sqrt{t}}\theta_{-\sqrt{t}}\psi_{\sqrt{t}}\theta_{\sqrt{t}}(p)$$

in a neighborhood of 0, forming a curve. Show that for $f \in C^\infty(M)$,

$$[X, Y]_p f = \lim_{t\to 0} \frac{f(\beta(t)) - f(\beta(0))}{t}.$$

Since our limit is over $t > 0$ (considering the domain of square roots), it is equal to

$$\lim_{t\to 0} \frac{f(\beta(t^2)) - f(p)}{t^2}.$$

Our goal is to use L'Hôpital's rule to compute this limit (by continuity the numerator tends to zero). Let us define

$$\gamma(x, y, z, w) = f(\psi_{-x}\theta_{-y}\psi_z\theta_w(p)),$$

so that $f(\beta(t^2)) = \gamma(t, t, t, t)$. So to compute the derivative of $f(\beta(t^2))$ we compute the partial derivatives of γ.

We have the following situation: we want to differentiate a function of the form

$$h(t) = f \circ \theta_t \circ g(p)$$

where θ_t is the flow of some vector field X and f is real-valued. Then

$$h'(t) = df_{\theta_t(gp)}(\theta^{gp})'(t),$$

and since θ is the flow of X, this is equal to

$$h'(t) = df_{\theta_t(gp)}X_{\theta_t(gp)} = X_{\theta_t(gp)}f.$$

In our case, this gives us

$$\begin{aligned}
\frac{\partial\gamma}{\partial x} &= -Y_{\psi_{-x}\theta_{-y}\psi_z\theta_w(p)}f \\
\frac{\partial\gamma}{\partial y} &= -X_{\theta_{-y}\psi_z\theta_w(p)}(f \circ \psi_{-x}) \\
\frac{\partial\gamma}{\partial z} &= Y_{\psi_z\theta_w(p)}(f\psi_{-x}\theta_{-y}) \\
\frac{\partial\gamma}{\partial w} &= X_{\theta_w(p)}(f\psi_{-x}\theta_{-y}\psi_z).
\end{aligned}$$

Thus

$$f(\beta(t^2))'(0) = \sum_i \frac{\partial\gamma}{\partial x_i}(0) = -Y_pf - X_pf + Y_pf + X_pf = 0,$$

and so we can apply L'Hôpital again (the denominator is $2t$). But we need to differentiate these partial derivatives. These are functions of the form

$$k(t) = X_{h\circ\theta_t\circ f}\, g = (Xg)(h \circ \theta_t \circ f(p)).$$

where X is a vector field and θ is the flow of some other vector field Y, g some real-valued smooth function, and f, h some smooth functions. Then

$$k'(t) = d(Xg)_{h\theta_t f(p)} dh_{\theta_t fp} (\theta^{fp})'(t) = d(Xg)_{h\theta_t fp} dh_{\theta_t fp} Y_{\theta_t fp} = Y((Xg) \circ h)_{\theta_t fp}.$$

So we have (by symmetry of second-order partial derivatives):

$$\begin{aligned}
\frac{\partial\gamma}{\partial xx} &= Y(Yf)_{\psi_{-x}\theta_{-y}\psi_z\theta_w(p)} \\
\frac{\partial\gamma}{\partial xy} &= X((Yf)\circ\psi_{-x})_{\theta_{-y}\psi_z\theta_w} \\
\frac{\partial\gamma}{\partial xz} &= -Y((Yf)\psi_{-x}\theta_{-y})_{\psi_z\theta_w} \\
\frac{\partial\gamma}{\partial xw} &= -X((Yf)\psi_{-x}\theta_{-y}\psi_z)_{\theta_w} \\
\frac{\partial\gamma}{\partial yy} &= X(X(f\psi_{-x}))_{\theta_{-y}\psi_z\theta_w} \\
\frac{\partial\gamma}{\partial yz} &= -Y(X(f\psi_{-x})\theta_{-y})_{\psi_z\theta_w} \\
\frac{\partial\gamma}{\partial yw} &= -X(X(f\psi_{-x})\theta_{-y}\psi_z)_{\theta_w} \\
\frac{\partial\gamma}{\partial zz} &= Y(Y(f\psi_{-x}\theta_{-y}))_{\psi_z\theta_w} \\
\frac{\partial\gamma}{\partial zw} &= X(Y(f\psi_{-x}\theta_{-y})\psi_z)_{\theta_w} \\
\frac{\partial\gamma}{\partial ww} &= X(X(f\psi_{-x}\theta_{-y}\psi_z))_{\theta_w}.
\end{aligned}$$

So

$$f(\beta(t^2))''(0) = \sum_{i,j} \frac{\partial\gamma}{\partial x_i x_j}(0),$$

and evaluating the second order partial derivatives at 0 gives results like Y_pYf, X_pYf, etc. Then dividing by 2 (the denominator of the limit is 2, the second derivative of t^2) gives that the limit is equal to

$$X_pYf - Y_pXf = [X,Y]_pf$$

as desired.

5.2 Problem

Prove that there is a unique smooth structure on $\mathbb{R}$ with its standard topology, up to diffeomorphism.

Let M be a smooth manifold over $\mathbb{R}$ with the same topology. Suppose that there is a nowhere vanishing vector field on M, call it X, and let $\gamma: I \to M$ be a maximal integral curve starting at 0. Then

$$\gamma'(t) = X_{\gamma(t)} \neq 0$$

means that $\gamma'(t)$ has full rank (one) always, and is thus a submersion and immersion. Furthermore integral curves are either periodic or injective; since γ has nonzero derivative it cannot be periodic, so it is injective.

Since γ is a submersion, it is an open map, so $\gamma(I) \subseteq M$ is open and connected. If γ is not surjective then $\gamma(I)$ is an open interval (a, b) with one side finite, assume b is. But then there is another integral curve of X starting at b, call it δ, and similarly its image in M is an open interval and thus intersects (a, b). Let $\delta(s) \in (a, b)$ so that $p = \gamma(t) = \delta(s)$ for some t, and then $\gamma(x + t)$ and $\delta(x + s)$ are both integral curves starting at p, so they agree in their common domain. But then we can extend γ to part of δ's domain, including b. This contradicts γ's maximality, so γ's image must be all of M.

Thus γ is bijective, an immersion, and a submersion. Thus it is a diffeomorphism, and since $I \subseteq \mathbb{R}$ is an open interval and thus diffeomorphic to $\mathbb{R}$, M and $\mathbb{R}$ are diffeomorphic.

Now we show that a nowhere vanishing X exists. Let us cover M by smooth charts on open intervals (a_i, b_i), and by reordering and shrinking we can assume $a_i < b_{i-1} < a_{i+1} < b_i$ for all i (so every point is contained in at most two open sets). Furthermore, we can assume that this atlas is oriented: we start at $i = 0$ and work forward swapping orientation of the charts if they aren't compatible with the orientation of (a_0, b_0)'s chart. Similarly we work backward.

Now define the vector field X^i on (a_i, b_i) to be given by d/dt (relative to the chart). If we let τ^i be the transition function between (a_i, b_i) and (a_{i+1}, b_{i+1}), then X^i_p relative to (a_{i+1}, b_{i+1}) is $(\tau_i)' d/dt$. Since the atlas is oriented, $(\tau_i)' > 0$.

Now let λ_i be a partition of unity subordinate to the open cover, and let

$$X = \sum_i \lambda_i X^i.$$

This is smooth by localy finiteness, and thus forms a vector field. Furthermore for $p \in M$, it is contained in a single (a_i, b_i) or two. If it is contained in a single one, then $X_p = \lambda_i(p) X^i_p = X^i_p$ (since the partition must sum to one), which is nonzero by definition. Otherwise,

$$X_p = \lambda_i(p) X^i_p + \lambda_{i+1}(p) X^{i+1}_p = \lambda_i(p) \left.\frac{d}{dt}\right|_p + (1 - \lambda_i(p)) \tau_i'(p) \left.\frac{d}{dt}\right|_p.$$

This cannot be zero, if it were then

$$\lambda_i(p)(\tau_i'(p) - 1) = \tau_i'(p),$$

But $\lambda_i(p)(\tau_i'(p) - 1) \leq \tau_i'(p) - 1 < \tau_i'(p)$ a contradiction.

5.3 Problem

Let $G = \mathrm{GL}_n(\mathbb{C})$ be the Lie group of complex invertible $n \times n$ matrices. Its lie group $\mathfrak{g} \cong T_e G$ can be identified with $\mathrm{Mat}_n(\mathbb{C})$.

(1) Prove an explicit formula for left-invariant vector fields on G.

(2) Prove an explicit formula for the exponential map $\exp\colon \mathfrak{g} \to G$.

(3) Let $X, Y \in \mathfrak{g}$, show in two ways that $[X, Y] = XY - YX$.

- **(i)** Use part (1) and the definition of Lie brackets.
- **(ii)** Prove an explicit formula for $\mathrm{Ad}\colon G \to \mathrm{Aut}(\mathfrak{g})$, use the fact that $\mathrm{ad} = \mathrm{Ad}_*$ and $\mathrm{ad}_X(Y) = [X, Y]$, as well as part (2).

(1) We know in general for $v \in T_e G$, it defines the left-invariant vector field $v^L \in \mathrm{Lie}(G)$ given by

$$v^L_g = d(L_g)_e v.$$

So we need to compute the differentials of left multiplication. Recall the identification $\mathrm{Mat}_n(\mathbb{C}) \leftrightarrow T_I G$ by mapping a matrix $(A^i_j) \in \mathrm{Mat}_n(\mathbb{C})$ to

$$(A^i_j) \leftrightarrow A^{i,b}_j \frac{\partial}{\partial x^{i,b}_j}$$

where $A^{i,0}_j = \mathrm{Re}(A^i_j)$ and $A^{i,1}_j = \mathrm{Im}(A^i_j)$.

For a matrix A recall that

$$d(L_A)_I \left(\left. \frac{\partial}{\partial x^{i,b}_j} \right|_I \right) = \frac{\partial (L_A)^{\ell,c}_k}{\partial x^{i,b}_j}(A) \left. \frac{\partial}{\partial x^{\ell,c}_k} \right|_A ,$$

so we compute the partial derivatives of L_A. Let $(X^{i,b}_j)$ be a complex-valued matrix,

$$(L_A X)^\ell_k = (AX)^\ell_k = A^\ell_t X^t_k,$$

And thus

$$(L_A X)^{\ell,0}_k = \mathrm{Re}((AX)^\ell_k) = (A^{\ell,0}_t X^{t,0}_k) - (A^{\ell,1}_t X^{t,1}_k)$$

so that

$$\frac{\partial (L_A)^{\ell,0}_k}{\partial x^{t,0}_k} = A^{\ell,0}_t, \qquad \frac{\partial (L_A)^{\ell,0}_k}{\partial x^{t,1}_k} = -A^{\ell,1}_t.$$

Similarly

$$(L_A X)^{\ell,1}_k = A^{\ell,0}_t X^{t,1}_k + A^{\ell,1}_t X^{t,0}_k$$

so that

$$\frac{\partial (L_A)^{\ell,1}_k}{\partial x^{t,0}_k} = A^{\ell,1}_t, \qquad \frac{\partial (L_A)^{\ell,1}_k}{\partial x^{t,1}_k} = A^{\ell,0}_t.$$

All other partial derivatives are zero.

So we have that

$$d(L_A)_I \left(\left. \frac{\partial}{\partial x^{i,0}_j} \right|_I \right) = A^{\ell,0}_i \left. \frac{\partial}{\partial x^{\ell,0}_j} \right|_A + A^{\ell,1}_i \left. \frac{\partial}{\partial x^{\ell,1}_j} \right|_A ,$$

and

$$d(L_A)_I \left(\left. \frac{\partial}{\partial x^{i,1}_j} \right|_I \right) = -A^{\ell,0}_i \left. \frac{\partial}{\partial x^{\ell,0}_j} \right|_A + A^{\ell,0}_i \left. \frac{\partial}{\partial x^{\ell,1}_j} \right|_A .$$

Under the correspondence $\partial/\partial x^{i,0}_j\Big|$ is the basis matrix E_{ij}, and $\partial/\partial x^{i,1}_j\Big| = \mathrm{i}E_{ij}$ (multiplication by the imaginary unit). We see then that

$$d(L_A)_I(E_{ij}) = AE_{ij}, \qquad d(L_A)_I(\mathrm{i}E_{ij}) = A\mathrm{i}E_{ij}$$

under the natural correspondence.

By linearity we see that

$$d(L_A)_I(B) = AB$$

for all matrices B. Thus the left-invariant vector field corresponding to $B \in T_I G$ is

$$B^L|_A = d(L_A)_I(B) = AB.$$

(2) For a matrix A define

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k.$$

By submultiplicity of the Frobenius norm, this converges absolutely for all A (as we can compare the series with $|A|^k/k!$ which converges to $e^{|A|}$). We claim that $\exp(A) = e^A$ for all matrices.

To do this, we need to show that for $A \in \mathfrak{g}$, its one-parameter subgroup is given by $\gamma(t) = e^{tA}$. We need to show that $\gamma(t)$ is invertible for all t (so that it lands in G), $\gamma'(t) = A^L|_{\gamma(t)}$ for all t (so it defines the one-parameter subgroup), and $\gamma(0) = I$. The last point is the easiest as it follows from the definition that $e^0 = I$.

For the second point, we need to show

$$\gamma'(t) = A^L|_{\gamma(t)} = \gamma(t)A$$

by our formula for left-invariant vector fields. Note that

$$\gamma(t) = \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$$

is a power series which converges uniformly in any closed interval, and thus we can differentiate it term by term:

$$\gamma'(t) = \sum_{k=1}^{\infty} \frac{t^{k-1}}{(k-1)!} A^k = \left(\sum_{k=0}^{\infty} \frac{t^k}{k!} A^k \right) A = \gamma(t)A$$

as desired.

Finally to show that $\gamma(t)$ is invertible, we will show that its inverse is $\gamma(-t)$ (not surprising by exponential laws). So define $\sigma(t) = \gamma(t)\gamma(-t)$, we will show that it has zero derivative. By the product rule and what we just showed,

$$\sigma'(t) = \gamma'(t)\gamma(-t) + \gamma(t)\gamma'(-t) = \gamma(t)A\gamma(-t) - \gamma(t)A\gamma(-t) = 0$$

so $\sigma(t)$ is constant. Since $\sigma(0) = \gamma(0)\gamma(0) = I$, we see that $\sigma(t) = I$ for all t and thus $\gamma(t)$ is invertible for all t. Thus $\gamma(t)$ is A^L's integral flow starting at I, thus the one-parameter subgroup generated by A. Then by definition

$$\exp(A) = \gamma(1) = e^A$$

as desired.

(3)

(i) We recall that

$$[A, B] = [A^L, B^L]_I$$

and for vector fields,

$$[A, B] = \left(A^i \frac{\partial B^j}{\partial x^i} - B^i \frac{\partial A^j}{\partial x^i} \right) \frac{\partial}{\partial x^j}.$$

In our case as we showed,

$$A^L_X = XA = X^i_k A^k_j \frac{\partial}{\partial x^i_j},$$

where we split the real and imaginary parts. Then

$$(A^L)^i_j(X) = X^i_k A^k_j \implies \frac{\partial (A^L)^i_j}{\partial x^i_k} = A^k_j$$

thus

$$[A^L, B^L]_I = \left((A^L)^i_j(I) B^j_k - (B^L)^i_j(I) A^j_k \right) \frac{\partial}{\partial x^i_k} = (A^i_j B^j_k - B^i_j A^j_k) \frac{\partial}{\partial x^i_k}$$

which is $AB - BA$ under the correspondence.

(ii) By naturality of exp, the following diagram commutes

$$\begin{tikzcd} \mathfrak{g} \arrow[r, "\mathrm{Ad}_*"'] \arrow[d, "\exp"'] & \mathrm{end}(\mathfrak{g}) \arrow[d, "\exp'"'] \\ G \arrow[r, "\mathrm{Ad}"] & \mathrm{Aut}(\mathfrak{g}) \end{tikzcd}$$

Since $\mathrm{Aut}(\mathfrak{g})$ is $\mathrm{GL}(\mathfrak{g})$, we already know that $\exp'$ is given by the exponential map defined above:

$$\exp'(\Phi) = \sum_k \frac{1}{k!}\Phi^k, \qquad \Phi \in \mathrm{end}(\mathfrak{g}),$$

so we write exp for $\exp'$ as well. Since $\mathrm{ad} = \mathrm{Ad}_*$ we have

$$\mathrm{Ad}(\exp(A)) = \exp(\mathrm{ad}(A)).$$

By linearity, $\mathrm{ad}(A)$'s one-parameter subgroup is given by $\gamma(t) = \exp(t\mathrm{ad}(A)) = \exp(\mathrm{ad}(tA)) = \mathrm{Ad}(\exp(tA))$. This means that $\gamma'(0) = \mathrm{ad}(A) = [A, \bullet]$, so we want to differentiate $\mathrm{Ad}(\exp(tA))$.

Now we claim that $\mathrm{Ad}(A)B = ABA^{-1}$ for all $A \in G, B \in \mathfrak{g}$. Indeed,

$$\mathrm{Ad}(A) = (R_{A^{-1}})_* \circ (L_A)_*$$

and we showed that $(L_A)_* = d(L_A)_I$ is left multiplication by A, dually $(R_A)_*$ is right multiplication by A. Thus $\mathrm{Ad}(A)B = ABA^{-1}$ as desired.

We know that

$$[A, B] = \mathrm{ad}(A)B = \left(\frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(\exp(tA))\right)B = \frac{d}{dt}\bigg|_{t=0} (\mathrm{Ad}(\exp(tA))B),$$

and since $\exp(A)^{-1} = \exp(-A)$,

$$\mathrm{Ad}(\exp(tA))B = \exp(tA)B\exp(-tA)$$

so that

$$\begin{aligned} \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(\exp(tA))B &= \left(\frac{d}{dt}\bigg|_{t=0} \exp(tA)\right)B\exp(0) + \exp(0)B\left(\frac{d}{dt}\bigg|_{t=0} \exp(-tA)\right) \\ &= AB - BA, \end{aligned}$$

where the last equality is because $\exp(tA)$ is the one-parameter subgroup generated by A. So we have shown

$$[A, B] = AB - BA$$

as desired.

5.4 Problem

Consider the Lie group $G = U_n$ of complex unitary $n \times n$ matrices and its Lie algebra $\mathfrak{u}_n$. In a previous homework we showed a natural identification $\mathfrak{u}_n \cong T_e U_n$ with the space of skew-Hermitian matrices ($M^\dagger = -M$).

(1) Repeat the previous problem for G.

(2) Define an inner product on $T_e U_n$ by

$$g_e(X,Y) = \operatorname{Re}\operatorname{tr}(X^\dagger Y).$$

Show that g_e extends to a bi-invariant metric g on U_n.

(3) Verify that $g([X,Y],Z) = -g(Y,[X,Z])$ for $X,Y,Z \in \mathfrak{u}_n$.

(1) U_n is an embedded Lie subgroup of $\mathrm{GL}_n(\mathbb{C})$ as we showed in the previous homework. Thus for $U \in U_n$, $L_U\colon U_n \to U_n$ is the restriction of $L_U\colon \mathrm{GL}_n(\mathbb{C}) \to \mathrm{GL}_n(\mathbb{C})$ and thus dL_U is given by the same formula as for $\mathrm{GL}_n(\mathbb{C})$. That is, left-multiplication by U. So left-invariant vector fields are of the form

$$M^L_U = UM, \qquad U \in U_n, M \in \mathfrak{u}_n.$$

Since U_n is a Lie subgroup, its exponential is the restriction of that of $\mathrm{GL}_n(\mathbb{C})$ so it is given by the same formula (matrix exponentiation), and the Lie bracket is as well.

(2) If we have an inner product $g_e\bullet,\bullet$ on $\mathfrak{g}$ then we showed it extends to a metric on G by

$$g(X,Y) = g_e(X_e, Y_e).$$

This is right-invariant when it is invariant under Ad. This is because for $X \in \mathfrak{u}_n$ and $U \in U_n$ we have $\mathrm{Ad}(U)X = (R_{U^{-1}})_*(L_U)_*X = (R_{U^{-1}})X$ by left-invariance. Thus for $X, Y \in \mathfrak{u}_n$,

$$\begin{aligned}((R_{U^{-1}})^*g)_V(X_V, Y_V) &= g_{VU^{-1}}\big(((R_{U^{-1}})_*X)_{VU^{-1}}, ((R_{U^{-1}})_*Y)_{VU^{-1}}\big)\\ &= g_{VU^{-1}}((\mathrm{Ad}(U)X)_{VU^{-1}}, (\mathrm{Ad}(U)Y)_{VU^{-1}}) = g_I(\mathrm{Ad}(U)X, \mathrm{Ad}(V)Y).\end{aligned}$$

The last equality is due to how we extended g_e: $g_U(X_U, Y_U) = g_e(X_e, Y_e)$. So if g_e is Ad-invariant then

$$((R_{U^{-1}})^*g)_V(X_V, Y_V) = g_I(X,Y) = g_V(X_V, Y_V),$$

so that g is right-invariant.

Again Ad on U_n is the restriction of Ad on $\mathrm{GL}_n(\mathbb{C})$ so $\mathrm{Ad}(U)X = UXU^{-1} = UXU^\dagger$ since U is unitary. Thus

$$g_I(\mathrm{Ad}(U)X, \mathrm{Ad}(U)Y) = g_I(UXU^\dagger, UYU^\dagger) = \operatorname{Re}\operatorname{tr}((UXU^\dagger)^\dagger(UYU^\dagger)) = \operatorname{Re}\operatorname{tr}(UX^\dagger YU^\dagger).$$

Permuting the matrices in the trace, this is equal to

$$= \operatorname{Re}\operatorname{tr}(U^\dagger UX^\dagger Y) = \operatorname{Re}\operatorname{tr}(X^\dagger Y) = g_I(X,Y).$$

So g_I is Ad-invariant as desired.

For completeness, g_I is an inner product because $(X,Y) \mapsto \operatorname{tr}(X^\dagger Y)$ is a complex inner product (the Frobenius inner product), and taking the real part of a complex inner product is a real inner product.

(3) We compute directly:

$$g_I([X,Y],Z) = \operatorname{Re}\operatorname{tr}((XY - YX)^\dagger Z) = \operatorname{Re}\operatorname{tr}(Y^\dagger X^\dagger Z) - \operatorname{Re}\operatorname{tr}(X^\dagger Y^\dagger Z),$$

since X, Y are skew-Hermitian this is equal to

$$g_I([X,Y],Z) = \operatorname{Re}\operatorname{tr}(YXZ) - \operatorname{Re}\operatorname{tr}(XYZ).$$

And

$$g_I(Y,[X,Z]) = \operatorname{Re}\operatorname{tr}(Y^\dagger(XZ-ZX)) = -\operatorname{Re}\operatorname{tr}(YXZ-YZX) = -(\operatorname{Re}\operatorname{tr}(YXZ) - \operatorname{Re}\operatorname{tr}(YZX))$$

permuting cyclicly the product in the last trace gives us

$$g_I(Y,[X,Z]) = -(\operatorname{Re}\operatorname{tr}(YXZ) - \operatorname{Re}\operatorname{tr}(XYZ)) = -g_I([X,Y],Z)$$

as desired.

5.5 Problem

Let $\mathrm{SL}_n(\mathbb{R})$ denote the group of $n \times n$ real matrices with unit determinant. We show that it is a Lie group of dimension n^2-1 with Lie algebra $\mathfrak{sl}_n$, $n \times n$ real matrices with zero trace.

We also define SU_n to be the group of $n \times n$ unitary matrices with unit determinant. We show that it is a Lie group with Lie algebra $\mathfrak{su}_n$, $n \times n$ skew-Hermitian matrices of trace 0.

(1) Let $F\colon \mathrm{Mat}_n(\mathbb{R}) \to \mathbb{R}$ be given by taking the determinant. Thus for $A \in \mathrm{Mat}_n(\mathbb{R})$, its differential $dF_A\colon \mathrm{Mat}_n(\mathbb{R}) \to \mathbb{R}$ is a linear map. Show that

$$dF_A(B) = \operatorname{tr}(\operatorname{adj}(A)B)$$

where $\operatorname{adj}(A)$ is the adjugate of A.

(2) Use the previous point to show that 1 is a regular value of det. Conclude that $\mathrm{SL}_n(\mathbb{R})$ is a smooth manifold of dimension n^2-1 and $T_2\mathrm{SL}_n(\mathbb{R}) = \mathfrak{sl}_n(\mathbb{R})$.

(3) Prove that group multiplication and inversion on $\mathrm{SL}_n(\mathbb{R})$ are smooth, so that $\mathrm{SL}_n(\mathbb{R})$ is a Lie group.

(4) Identify S^1 with the unit circle in $\mathbb{C}$, so a Lie group. Define the group morphism $h\colon U_n \to S^1$ by $h(U) = \det U$. Show that h is smooth and $\mathrm{id} \in U_n$ is a regular point of h.

(5) Use the fact that h is a homomorphism to show that it is a submersion.

(6) Conclude that SU_n is a Lie group.

(7) Let $\iota\colon U_n \hookrightarrow \mathrm{Mat}_n(\mathbb{C})$ be the inclusion. Show that the composition

$$T_I U_n \xrightarrow{d\iota_I} T_I\mathrm{Mat}_n(\mathbb{C}) \xrightarrow{\zeta} \mathrm{Mat}_n(\mathbb{C})$$

carries $T_I\mathrm{SU}_n \subseteq T_I U_n$ to $\mathfrak{su}_n \subseteq \mathfrak{u}_n$, where ζ maps $[\gamma]$ to $\gamma'(0)$.

(1) We know that

$$dF_A\left(\left.\frac{\partial}{\partial x^i_j}\right|_A\right) = \frac{\partial F}{\partial x^i_j}(A)\frac{d}{dt}.$$

$\partial/\partial x^i_j\big|_A$ corresponds to the standard basis matrix E^i_j (zero except at the (i,j) coordinate), so we want to show that this is equal to

$$dF_A(E^i_j) = \frac{\partial F}{\partial x^i_j}(A)\frac{d}{dt} = \operatorname{tr}(\operatorname{adj}(A)E^i_j)\frac{d}{dt},$$

and then by linearity we get the result for all B, not just E^i_j. So we want to show that

$$\frac{\partial F}{\partial x^i_j}(A) = \operatorname{tr}(\operatorname{adj}(A)E^i_j) = (\operatorname{adj}(A)E^i_j)^k_k = \operatorname{adj}(A)^j_i = (-1)^{i+j}\det(A_{ij})$$

where A_{ij} is the minor obtained by removing the ith row and jth column of A.

For $X = (x^i_j)$,

$$F(X) = \sum_{\sigma\in S_n} \operatorname{sgn}\sigma \prod_{i=1}^n x^i_{\sigma(i)},$$

so

$$\frac{\partial F}{\partial x^i_j} = \sum_{\sigma(i)=j} \operatorname{sgn}\sigma \prod_{k\neq i} x^k_{\sigma(k)}.$$

Notice that

$$\frac{\partial F}{\partial x^i_j}(A) = \sum_{\sigma(i)=j} \operatorname{sgn}\sigma \prod_{k\neq i} a^k_{\sigma(k)} = (-1)^{i+j}\det(A_{ij})$$

as desired, since the permutations run over $\sigma \in S_{n-1}$ (identified with the subset of S_n where $\sigma(i) = j$, and this identification changes the sign by $(1)^{i+j}$).

(2) If $F(A) = 1$, meaning $\det(A) = 1$, then A is invertible and $\operatorname{adj}(A) = A^{-1}$. Thus

$$dF_A(A) = \operatorname{tr}(\operatorname{adj}(A)A) = \operatorname{tr}(I) = n \neq 0.$$

This means that dF_A is a nonzero linear functional, and thus surjective (as its codomain is one-dimensional). In other words, A is a regular point of F, so 1 is a regular value.

Since $\mathrm{SL}_n(\mathbb{R}) = F^{-1}(1)$, by the regular level set theorem this means $\mathrm{SL}_n(\mathbb{R})$ is an embedded smooth submanifold of $\mathrm{Mat}_n(\mathbb{R})$, whose codimension is $\dim\mathbb{R} = 1$. Thus its dimension is $\dim \mathrm{SL}_n(\mathbb{R}) = \dim \mathrm{Mat}_n(\mathbb{R}) - 1 = n^2 - 1$.

Finally we also know that $T_I\mathrm{SL}_n(\mathbb{R}) = \ker dF_I$. This is because for a regular level set $F^{-1}q$, its dimension equals $\ker dF_p$ (since dF_p is surjective, the kernel has codimension equal to the dimension of its codomain), and F is constant on $F^{-1}q$ so $dF_p = 0$ on the level set, so $T_pF^{-1}q$ is contained in the kernel. Their dimensions are equal, so they are equal.

Since $dF_I(B) = \operatorname{tr}(\operatorname{adj}(I)^{-1}B) = \operatorname{tr}(B)$, its kernel is all the matrices B of zero trace. Thus $T_I\mathrm{SL}_n(\mathbb{R}) =_n (\mathbb{R})$ as desired.

(3) $\mathrm{SL}_n(\mathbb{R})$ is a subgroup of $\mathrm{GL}_n(\mathbb{R})$ and an embedded submanifold of $\mathrm{Mat}_n(\mathbb{R})$. Thus it is embedded in $\mathrm{GL}_n(\mathbb{R})$ as well, and multiplication and inversion in $\mathrm{GL}_n(\mathbb{R})$ restrict to $\mathrm{SL}_n(\mathbb{R})$. We showed in a previous homework that we can restrict both the domain and codomain of a smooth function to embedded submanifolds, so doing this for multiplication and inversion (restricting the smooth maps of $\mathrm{GL}_n(\mathbb{R})$ to $\mathrm{SL}_n(\mathbb{R})$) shows that multiplication and inversion of $\mathrm{SL}_n(\mathbb{R})$ is smooth.

(4) S^1 is embedded in $\mathbb{C}$, and U_n in $\mathrm{Mat}_n(\mathbb{C})$. The map $\det\colon \mathrm{Mat}_n(\mathbb{C}) \to \mathbb{C}$ is smooth, and we can restrict it to the embedded submanifolds (on both the domain and codomain) $U_n \subseteq \mathrm{Mat}_n(\mathbb{C})$ and $S^1 \subseteq \mathbb{C}$. Thus $\det\colon U_n \to S^1$ is smooth.

A similar computation as in part (1) shows that

$$dh_I(A) = \operatorname{tr}(\operatorname{adj}(I)^{-1}A) = \operatorname{tr}(A).$$

Since $T_IU_n \cong \mathfrak{u}_n$, the space of matrices skew-Hermitian matrices, taking $A = iI$ we see that $dh_I(A)$ is nonzero. Since S^1 is one-dimensional, this means that dh_I has full-rank and thus I is a regular point.

(5) h is a group homomorphism between Lie groups (by multiplicity of the determinant), and thus has constant rank. Since h has rank 1 at I, it has rank 1 everywhere. Since $\dim S^1 = 1$, this means that h is a submersion.

(6) $\mathrm{SU}_n = h^{-1}(1)$ is a regular level set (since h is a submersion, every level set is regular), and thus an embedded submanifold of U_n. It is also a subgroup of U_n, and since we can restrict smooth functions to embedded submanifolds, restricting multiplication and inversion of U_n to SU_n shows that they are smooth, so it is a Lie group.

(7) SU_n's codimension is equal to the rank of h, so 1. Thus

$$\dim \mathrm{SU}_n = \dim U_n - 1 = n^2 - 1.$$

The dimension of $\mathfrak{su}_n$ is also $n^2 - 1$: a skew-Hermitian matrix is determined by its values on its upper triangle, with imaginary values on its diagonal. Requiring that its trace be zero gives us $n-1$ choices of the diagonal, so

$$\dim \mathfrak{su}_n = \frac{n^2 - n}{2} \cdot 2 + (n-1) = n^2 - 1,$$

where we multiply the first term by 2 since each element of the upper triangle is complex (real + imaginary parts). Explicitly, a basis for $\mathfrak{su}_n$ is given by

$$\left\{ E^i_j - E^j_i, \mathrm{i}E^i_j + \mathrm{i}E^i_j \;\middle|\; i > j \right\} \cup \left\{ \mathrm{i}E^i_i - \mathrm{i}E^n_n \;\middle|\; i < n \right\}.$$

Since $\zeta \circ d\iota_I$ is a composition of injections, it is an injection ($d\iota_I$ is an injection since U_n is embedded and thus ι is an immersion). So we need only show that it carries $T_I\mathrm{SU}_n$ into $\mathfrak{su}_n$, and then considering dimensions this shows that the image of $T_I\mathrm{SU}_n$ is equal to $\mathfrak{su}_n$.

Let $\gamma\colon (-\varepsilon, \varepsilon) \to \mathrm{SU}_n$ be a curve starting at I. Then

$$\zeta \circ d\iota_I[\gamma] = \zeta[\iota \circ \gamma] = (\iota \circ \gamma)'(0).$$

So we identify γ with $\iota \circ \gamma$ and view it as a curve into $\mathrm{Mat}_n(\mathbb{C})$ with values in SU_n (meaning $\gamma(t)$ is unitary with unit determinant for all t). We already showed that $\gamma'(0) \in \mathfrak{u}_n$ in a previous homework, so we need only show that $\mathrm{tr}\gamma'(0) = 0$.

We know that $\det \gamma(t) = 1$, and thus

$$0 = \left.\frac{d}{dt}\right|_{t=0} \det \gamma(t) = d\det\nolimits_{\gamma(0)} \gamma'(0) = d\det\nolimits_I(\gamma'(0)).$$

We showed above that this is equal to

$$= \mathrm{tr}(\mathrm{adj}(I)\gamma'(0)) = \mathrm{tr}(\gamma'(0)),$$

so $\gamma'(0)$ has trace zero, as desired.